

MBML Revision Cheat Sheet

1. Probabilistic Foundations

Sum Rule (Marginalization): $p(x) = \sum_y p(x, y)$

Product Rule (Chain Rule): $p(x, y) = p(y|x)p(x)$

Misleading Identities: $p(A|B) = \sum_c p(A, C | B)$ is true, but: $p(A|B) = \sum_c p(A | B, C)$ is not true, because $p(A|B)$ is not necessarily equal to $p(A|B, C)$ for all C .

Bayes' Theorem:

$$p(\theta | D) = \frac{p(D|\theta)p(\theta)}{p(D)}$$

- $p(\theta|D)$: Posterior
- $p(D|\theta)$: Likelihood
- $p(\theta)$: Prior
- $p(D)$: Evidence (Marginal Likelihood)

2. PGM Representation

- **Nodes:** Represent Random Variables.
 - **Shaded:** Observed data.
 - **Unshaded:** Latent/Hidden variables.
- **Edges:** Probabilistic dependencies.
- **Plates:** Denote N independent repetitions of the enclosed variables.
- **Joint Factorization:** A PGM defines the joint distribution as the product of all nodes conditioned on their parents:

$$p(x_1, \dots, x_n) = \prod_{i=1}^n p(x_i | \text{parents}(x_i))$$

3. Independence & D-Separation

$x \perp\!\!\!\perp y \mid z$ means x is independent of y given z .

- **Tail-to-Tail:** $x \leftarrow z \rightarrow y$. Path blocked if z is observed.
- **Head-to-Tail:** $x \rightarrow z \rightarrow y$. Path blocked if z is observed.
- **Head-to-Head (Collider):** $x \rightarrow z \leftarrow y$. Path **open** if z (or its descendants) is observed. This is “explaining away.”

4. Frequentist vs. Bayesian

- **Frequentist:** Parameters θ are fixed. We find θ_{MLE} by maximizing $p(D|\theta)$.
- **Bayesian:** Parameters are Random Variables with distributions. We seek the full posterior $p(\theta|D)$.
- **Conjugate Priors:** If the prior and posterior are in the same family (e.g., Beta prior + Bernoulli likelihood = Beta posterior), inference is exact and easy.

2.1 D-separation algorithm

For any expression “is **x** independent of **y** given **z**” (formally, $x \perp\!\!\!\perp y \mid z$):

1. Draw the ancestral graph:
 - It is the part of the original graph that has only the variable sets **x**, **y** and **z**, and all their ancestors among them.
2. Moralize the graph by marrying the parents.
 - For each pair of variables with a common child, draw an undirected edge between them. (if a variables has more than one parent, connect all parents with undirected edges).
3. Disorient the graph by replacing all edges for undirected edges.
4. Delete the variables **z** (and any other observed variables not explicitly included in **z**), and their edges.

Analysis of the result:

- if **x** and **y** are **disconnected**, then they are unconditionally independent given **z**.
 - Being disconnected means that there is no path between **x** and **y** in the resulting graph.
- Otherwise, they are not proven to be independent.

5. Regression Models

- **Linear-Gaussian:** $y \sim \text{Normal}(\beta^T x, \sigma^2)$. Assumes constant noise (homoscedastic).
- **Poisson Regression:** Used for count data (e.g., taxi pickups). $y \sim \text{Poisson}(\lambda)$ where $\lambda = \exp(\beta^T x)$.
- **Heteroscedastic:** Noise σ^2 is not constant but depends on input x : $\sigma^2(x)$.

6. Classification

- **Logistic Regression:** For binary $y \in \{0, 1\}$.

$$p(y = 1 \mid x) = \sigma(\beta^T x) = \frac{1}{1 + e^{-\beta^T x}}$$

- **Categorical:** Extension to K classes using the Softmax function.
- **Hierarchical Models:** Parameters vary by group (e.g., individual habits in travel data).

7. Mixture Models (Latent Variables)

Used for clustering or modeling multi-modal data.

- **Generative Story:**

1. Draw mixing proportions $\pi \sim \text{Dirichlet}(\alpha)$.
2. For each component k , draw parameters θ_k .
3. For each data point n :
 - Draw assignment $z_n \sim \text{Categorical}(\pi)$.
 - Draw data $x_n \sim p(x \mid \theta_{z_n})$.

8. Probabilistic Programming (Pyro)

- `pyro.sample("name", dist, obs=data)`: Defines a random variable. If `obs` is provided, it's an observed node.
- `pyro.plate("name", size)`: Creates a plate for vectorized operations.
- **Inference:** Translates the model into an algorithm (MCMC or Variational Inference) to find the posterior.

Exam Tip: When asked for a “Generative Story,” always write it as a step-by-step sampling process. Furthermore, when asked to draw the corresponding PGM. Even if the “Generative Story” describes specific distributions, the PGM should only include the random variables and their dependencies, without specifying the exact distributions. Thereby, keeping it more abstract and general.

8. Temporal Models

- **Markov Assumption:** $p(y_t \mid y_{1:t-1}) = p(y_t \mid y_{t-1})$. Future independent of past given present.
- **AR(M):** $y_t = \sum_{i=1}^M \beta_i y_{t-i} + \varepsilon$. Direct observations; noise accumulates in state.
- **SSM / LDS:** Separates latent “truth” (h_t) from noisy observations (y_t).
 - **Trans:** $h_t = A h_{t-1} + \eta_t$, $\eta_t \sim \mathcal{N}(0, R)$
 - **Emiss:** $y_t = C h_t + \varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$
- **Inference Types:** **Filtering** $p(h_t \mid y_{1:t})$ (live), **Smoothing** $p(h_t \mid y_{1:T})$ (post-hoc), **Prediction** $p(y_{t+k} \mid y_{1:t})$ (forecast).
- **HMM:** Discrete latent states h_t . Uses **Forward-Backward** or **Viterbi** (most likely sequence).
- **Extensions:** **EKF** (Non-linear via Taylor expansion), **SLDS** (Discrete z_t switches between linear regimes).

Model	Latent h_t	Key Tool
AR(M)	None	MLE / OLS
LDS / SSM	Continuous	Kalman Filter
HMM	Discrete	Forward-Backward
EKF	Non-linear	Linearization